

Project 06: Feed the Class

Unit: Unit 6, From Farm to Table (Food Systems and Production) **Topic:** 6.3 Feed the Class project **Runs across:** Lessons 6.12 (launch, constraints, menu choice), 6.13 (scale and cost, two days), 6.14 (flowchart and station card), 6.15 (consolidated shopping list and approval), 6.16 (cook day, two days), 6.17 (serve, eat, clean, reflect). Eight class periods, days 158 to 165 of the year. **Team:** the station team, about seven students at four stations, or nine to ten at three stations **Scored with:** Rubric 06 - Feed the Class.md (five criteria, 20 points; labs and projects category). The cook day lab score is taken separately on Rubric 01 - Lab Rubric.md (16 points), which is the same lab rubric used for every lab all year.

Standards this project assesses

- NYS FSP 1. Basic Culinary Skills a) Apply basic culinary and management skills to the preparation of nutritious food; c) Demonstrate safety and sanitation procedures when handling food and tools; d) Demonstrate accurate measuring of dry/solid, liquid, and small quantity ingredients; e) Prepare a simple recipe
- NYS FSP 2. Food Preparation e) Practice safety and sanitation practices and procedures in the preparation of food
- NYS FSP 5. Consumer Resources and Finance a) Demonstrate management of individual and family resources to select and purchase food; c) Apply consumer skills to selecting and purchasing of food items; d) Apply time management, organizational, and process skills to prioritize tasks and achieve goals
- NYS N&W 3. Consumer Resources and Finance a) Create and organize a shopping list (first taught in Unit 1, applied here at class scale)
- NYS N&W 1. Culinary Management g) Prepare and serve a simple meal; i) Practice food sharing behaviors that show consideration for other cultures and promote social acceptance
- CTE FCL 4. Buying Goods and Services c) Develop a spending and savings plan (budget) based on income and expenses; f) Apply a decision-making model to maximize consumer satisfaction when purchasing goods and services; g) Compare local, national, and global resources for obtaining goods and services; i) Demonstrate personal development of consumer skills through practice of these skills in a variety of classroom applications
- CTE HSW 4. Classroom and Workplace Safety c) Apply safety practices when using tools, machines, and equipment; e) Participate in housekeeping procedures that help to establish and maintain safe working conditions in the classroom or workplace
- CTE CIR 3. Workplace Communication e) Explain how to give and receive accurate reports in a variety of formats; f) Demonstrate effective communication skills in a group setting to accomplish a task
- CTE PSI 2. Design Process (Proactive) a) c. Defining constraints that must be adhered to
- CTE SUS 1. Resources e) Investigate practices that promote stewardship of environmental resources
- NYS FACS Standard 1 (Intermediate), Key Idea: Students will use an understanding of the elements of good nutrition to plan appropriate diets for themselves and others. They will know and use the appropriate tools and technologies for safe and healthy food

preparation.; PI: Students apply principles of food safety and sanitation.; PI: Students understand the relationships among diet, health, and physical activities; evaluate their own eating patterns; and use appropriate technology and resources to make food selections and prepare simple, nutritious meals.

- NYS FACS Standard 3 (Intermediate), Key Idea: Students will understand and be able to manage personal resources of talent, time, energy, and money and make effective decisions in order to balance their obligations to work, family, and self.; PI: Students are able to budget their time and money.; PI: Students understand the resources available to them, make informed decisions about the use of those resources, and know some ways to expand resources.
- CDOS 2 Integrated Learning; CDOS 3a Universal Foundation Skills: Managing Resources; Basic Skills; Managing Information; Personal Qualities; Interpersonal Skills
- National FCS 3.0: 2.1.1 Use time management, organization, and step-by-step process skills to rank tasks and reach goals.

The project sheet (student version)

Name: _____ Date: _____ Station: _____ Role: _____

The scenario

There are twenty-eight people in this room and I have about eighty dollars. In eight class periods, you are going to feed everybody one real meal, and I am not going to plan it, price it, or cook it for you. You are the kitchen.

Your station picks a dish. You scale the recipe from four servings up to the whole class. You price it, to the penny, from this week's real supermarket circular. You write your own flowchart in five-minute blocks and your own station card. You hand me one shopping list for the whole class and I buy exactly what is on it. Then you cook it, serve it to everybody else before you serve yourself, clean the room to a signed checklist, and tell me what it actually cost and what you would change.

I have done this before. Four kitchens in one period, 112 students across four periods on a \$150 budget, every price checked against the store circular. It worked because the plan was on paper before anybody touched a burner. That is the part you own.

The budget

- **\$20.00 per station.** That is what your station gets to spend on its own dish, for 28 portions plus two spares.
- **\$80.00 for the class.** Four stations, \$20 each. If your station comes in under, that is a good result, not a missed chance. Coming in under is what a real kitchen tries to do.
- **Cost per person** is the number you will report twice: what you planned in Lesson 6.15, and what the receipt actually said in Lesson 6.17.

All prices in the handouts are real Long Island supermarket prices to be refreshed from the current circular before teaching. The math works no matter what the prices are; only the answers change.

The constraints (the constraint card, and none of these are negotiable)

#	Constraint	What it rules out
1	No knives. Butter knives and kitchen shears only, and only if your card says so	Chopping, slicing, dicing. Anything that needs cutting comes pre-cut, pre-shredded, or is torn or pulled apart with two forks
2	No frying	Cutlets, anything in a pan of oil, anything in a fryer
3	Pre-cooked meat only, reheated to 165 F, or no meat	Raw chicken, raw ground beef, raw fish, raw eggs as an ingredient you handle
4	Stovetop, oven, and	Grills, blenders, food

	microwave only	processors, anything the room does not own
5	The posted class allergy list	Every item on the list, unless you write the substitution on your card
6	40 minutes, and the class has to eat inside the period	Anything that bakes for 45 minutes. Anything that has to rest for an hour
7	\$20.00 per station	Steak, shrimp, fresh berries in May, anything you cannot cover for 30 portions

The menu options

Five costed options, all priced for 28 portions at station scale. The class menu needs at least one main, at least one side or vegetable, and one dessert, so no two stations take the same option. Full cards with ingredients, prices, and instructions are on Handout 06.12 - Menu Option Cards and Constraint Card.md.

Option	Dish	Type	Station cost for 28 portions	Per person	Difficulty
A	Spaghetti with Marinara and Parmesan	Main	\$18.31	\$0.65	Medium. Boiling water, the teacher drains, the timing is tight
B	Pizza Tortillas	Main	\$21.32 with pepperoni, \$16.33 without	\$0.76 or \$0.58	Easy to cook, but the pepperoni line puts it \$1.32 over the station budget. Whoever takes it owes a fix
C	Bean and Cheese Burritos	Main	\$19.42	\$0.69	Easy. Stovetop and assembly, no meat
D	Garlic Bread and	Side	\$18.93	\$0.68	Easy. Oven and a butter

	Vegetable Tray				knife, and this station has support duty
E	No-Bake Cookie Dessert Cups	Dessert	\$18.94	\$0.68	Easiest. No heat at all

How a team picks. In Lesson 6.12 your station walks all five cards, fills the comparison table, then ranks its top two on a slip. I resolve conflicts by the rankings and by which station can handle the equipment. Then you write the three-sentence justification: which option you picked, and how it clears three of the seven constraints. If two stations both want the oven, the flowchart in Lesson 6.14 is where you solve it, not here.

What your team turns in

Everything goes in one project folder, handed in at the end of Lesson 6.17.

#	Deliverable	What it is	Due	Rubric criterion
1	The scaled recipe with the math shown	The scaling table: every ingredient, the recipe amount for 4, the scale factor, the scaled amount, the unit the store sells in, the packages to buy. Plus the unit conversion strip and the yield check showing the plan feeds at least 30	Lesson 6.13 Day 1	1. The scaling math and the yield
2	The costed shopping list	Your station's costing sheet, every line priced from the circular with a size, a unit price, a line cost, and a running total at or under \$20.00; the	Lesson 6.13 Day 2 and Lesson 6.15	2. The cost and the shopping list against the budget

		swap log with what each swap saved and what it cost in quality; plus your initials on the class consolidated list		
3	The station flowchart	Your poster: eight five-minute blocks from setup to clean up, a role name on every line, the critical path circled, and the actual times written in after cook day	Lesson 6.14, completed Lesson 6.16	3. The flowchart and station card
4	The station card	Ingredients for the station, ten numbered steps with one action each, three safety reminders plus one specific to your dish, the allergen box, and the four “what does done look like” checks	Lesson 6.14	3. The flowchart and station card
5	The cooked meal	The dish prepared, portioned, and served to the whole class inside the period, with every safety rule kept, every role done, and a signed station check	Lesson 6.16	4. The cook day
6	The reflection	Four questions	Lesson 6.17	5. The reflection

		answered in writing: what the plan said, what happened, what it cost per person (planned and actual), and what you would change, naming a block, a line, or a step		
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Also in the folder, not separately scored: the constraint card, the choice sheet with the justification, the role sheet, the pre-flight check, your duty strip, and the station check sheets.

Day by day

Day	Lesson	What happens	Checkpoint
1	6.12 (day 158)	The guess-the-total hook. The brief. The constraint card. The five-option walk. The station picks and writes its justification. Roles assigned	Choice sheet and role sheet handed in
2	6.13 Day 1 (day 159)	The scale factor. Scale every line. Watch the units: cups, fluid ounces, and ounces by weight. Round up to purchasable packages. The yield check against 30	Yield check clears 30
3	6.13 Day 2 (day 160)	Cost every line from the circular with a running total. The four totals go up on the wall. Revise with store-brand swaps and log them. Peer check, then the teacher signs	Teacher signs the costing sheet
4	6.14 (day 161)	Steps onto sticky notes, sticky notes	Station card passes the paired read;

		into eight blocks, a role name on every line. Circle the critical path. Write the station card. Paired read. Post to the class master grid and fix the collisions in red	poster on the wall
5	6.15 (day 162)	Read all four costing sheets aloud and combine duplicates. Build one class list sorted by store section. Unit price check and the swap column. The twelve-item approval checklist. Teacher signs and shops	Approved, signed consolidated list
6	6.16 Day 1 (day 163)	Pre-flight, six-step sanitation. Stations 1 and 2 cook and serve; Stations 3 and 4 prep, run service, run the clean check. Plates down at minute 29. Eat. Four station checks signed	Station check signed; actual times recorded
7	6.16 Day 2 (day 164)	Same, reversed. Stations 3 and 4 cook and serve; Stations 1 and 2 prep the final course, run service, run the clean check	Station check signed; actual times recorded
8	6.17 (day 165)	Set 30 covers by the Lesson 6.11 diagram. Read the allergen box. Serve others first. Eat together. Clean by four zones and count the waste. Write the four-question reflection. Folder in	Folder complete and handed in; Rubric 06 scored

Rubric 06 comes back at the start of Lesson 6.18 Day 2, before the review game and the test.

Team roles

Project roles (Lessons 6.12 to 6.15 and 6.17). Seven roles, one per student at a four-station class. At nine or ten students, add a Second Cost Lead and a Second Flow Lead.

Role	What you own
Station Manager	Runs the team. Owns the constraint card and the allergen box. Reads the allergy list at pre-flight and reads the allergen box to the class before service. Calls the critical path at minute 10 on cook day
Menu Lead	Owns the choice sheet and the justification. Knows why this dish and can say it in three sentences
Scale Lead	Owns the scaling table and the yield check. Reads every scaled line aloud to the team before it is written
Cost Lead	Owns the costing sheet and the running total. Writes a running total after every line, not at the end
Flow Lead	Owns the poster flowchart and the station card. Circles the critical path. Writes the ten steps, one action each
Shopping Lead	Owns the station's lines on the class consolidated list. Initials the rows that came from this station. Runs the unit price check
Reflection Lead	Owns the reflection. Collects the actual times from the Time Keeper and the receipt numbers, and makes sure all four questions get answered with evidence

Lab roles (Lesson 6.16 cook day, and Lesson 6.17 service). These are the room's standard role names, the same ones used in every lab all year. At seven students: Station Manager, Head Cook, Second Cook, Prep Cook, Helper Cook, Plate Builder, Time Keeper. At eight, add Toppings Chief. Nobody has to hold the same job in the planning and in the lab, and everybody holds a job on both cook days, either cooking or prepping and serving.

Where you get to choose

- **The dish.** Five options, and your station's rankings decide. A station may also propose a sixth option of its own, priced from the circular, for teacher approval.
- **The swaps.** Any store-brand or size swap that saves at least twenty-five cents and does not break the allergy list or the station card is yours to make.

- **The flowchart.** Nobody hands you the schedule. You build it, and you decide which shortcut to take if you run late.
- **The station card format.** Written or drawn. A picture flowchart card with eight boxes counts as a real card if a substitute could follow it.
- **The reflection format.** Written, typed, dictated to a teammate, recorded as a voice note with the numbers on paper, or drawn as a marked-up flowchart. Four questions, any form.
- **The table.** In Lesson 6.17 the host's opening sentence at your table can be in any language you speak.

ELL and IEP supports

The picture flowchart. Page 4 of Handout 06.14 is eight empty boxes, one per five-minute block, with room to draw the step and write one word under it. A team may submit this as its real station card, and a student may keep this version at their station on cook day even if the team submitted the written one. The eight blocks are also drawn as symbols: a pot, a clock, a flame, an oven, a checkmark, a plate, a fork, a sponge.

Sentence starters (Spanish, to be checked by a native speaker, a bilingual colleague, or the ENL teacher). The full set, with the reflection and the lab role starters, is in Vocabulary 06.md.

English	Spanish
We picked ___ because it fits ___.	Elegimos ___ porque cumple con ___.
The recipe makes 4. We need 28, so we multiply by 7.	La receta rinde 4. Necesitamos 28, así que multiplicamos por 7.
We will buy ___ instead of ___ and save \$___.	Vamos a comprar ___ en lugar de ___ y ahorrar \$___.
Our first step is . It takes minutes.	Nuestro primer paso es . Toma minutos.
Everything waits on , so it starts at minute .	Todo depende de , así que empieza en el minuto .
My job in this block is ___.	Mi trabajo en este bloque es ___.
Our plan said . What happened was .	Nuestro plan decía . Lo que pasó fue .
It cost \$___ per person. Next time I would change ___.	Costó \$___ por persona. La próxima vez cambiaría ___.

Other supports built in:

- A calculator for every student on every day of this project, not on request.
- Page 3 of Handout 06.13 is the scaling table with the scale factor already entered; page 5 is a four-page circular excerpt with only this project's items, for any student who finds a full circular overwhelming.
- The costing sheet and the consolidated list have the first two rows worked as examples.
- Every lab role has a fixed physical position, written on the duty strip and in the student's pocket, so a student who needs to know where to stand knows before the bell.

- Plate Builder, Time Keeper, and the prep station are the roles with no burner and no oven, and they are available to any student who needs them on both cook days, scored on the same rubric rows.
- The reflection may be dictated, recorded, or drawn.
- Extended time carries from Lesson 6.13 into 6.14 and from 6.17 into the start of 6.18. Cook day cannot be extended, so a student who needs more time has their tasks front-loaded into the early blocks instead.
- A student absent on a cook day does the written lab alternative under the grading plan (a step-by-step analysis of their own station card: what each step does, what could go wrong, and what done looks like), scored on Rubric 01 Objectives 1 to 3 out of 12 and scaled to 16, and is still scored on Rubric 06 criteria 1, 2, 3, and 5.

How it is scored

Rubric 06 - Feed the Class.md, five criteria, each Not Yet (1), Approaching (2), Meets (3), or Exceeds (4), for 20 points. Your team fills the self-assessment row before I score. Criteria 1, 2, 3, and 5 are team scores; criterion 4 (the cook day) has a team part and an individual part. A safety violation on cook day is an individual zero for that day's participation under the grading plan, is a conversation and a redo of the clean phase rather than a lecture, and caps the team at Approaching on criterion 4 for that day.

The cook day itself is also scored individually on Rubric 01 - Lab Rubric.md (16 points), which is the lab rubric used for every lab this year. You have both rubrics in hand from day one of the project. Nothing on either one should be a surprise.

The project counts in the labs and projects category, 40 percent of the marking period. A late project is accepted with a one-level drop on one criterion, never a zero.

The no-kitchen version

Everything in this project except the heat is unchanged. If the room has no kitchen, or the stove is down, or the class is in a regular classroom, the meal becomes a no-cook assembled meal and the project runs exactly the same eight days with the same six deliverables, the same rubric, and the same budget.

What changes: the five menu options become five assembly options, the flowchart phases become setup, portion, assemble, serve and eat, clean, and the critical path becomes whichever item has to be opened, drained, brought from a cooler, or set out first. Nothing else changes. Scaling, unit conversion, the yield check, costing from the circular, the swap log, the consolidated list, the twelve-item approval checklist, the station card, the pre-flight, the duty list, the serve-others-first rule, the four clean-up zones, the station check, and the reflection all stay.

The five no-cook options, priced for 28 portions at station scale (Long Island prices):

Option	Dish	Type	Station cost for 28	Per person	Notes
N1	Build-your-own wrap bar: flour tortillas, deli-sliced turkey or ham, pre-shredded lettuce, shredded cheese, mayonnaise and mustard packets	Main	2 packs tortillas \$6.98, 2 lb pre-sliced deli turkey about \$11.98, 1 bag shredded lettuce \$2.49 = \$21.45	\$0.77	Over by \$1.45. Swap to 1.5 lb deli meat and add a second cheese to make \$19.46. Cold items stay in a cooler with ice packs until minute 5
N2	Snack board: crackers, cheese sticks, baby carrots, grape tomatoes, hummus cups, pretzels	Main or side	Crackers 2 boxes \$5.98, cheese sticks 2 packs \$7.98, baby carrots 2 bags \$3.98 = \$17.94	\$0.64	Easiest. No knife, no heat, nothing to drain
N3	Overnight oats cups: rolled oats, milk or a milk alternative,	Main or breakfast	Oats 1 canister \$4.29, shelf-stable milk 2 quarts \$5.98, raisins 1 box	\$0.63	Assembled one day, eaten the next, which is a real hold-time and

	cinnamon, raisins or diced canned fruit, honey		\$3.49, cups 1 pack \$3.99 = \$17.75		refrigeration lesson
N4	Fresh salsa and chips with pre-cut vegetables and canned beans	Side	Tortilla chips 2 bags \$5.98, jarred salsa 2 jars \$4.98, canned black beans 3 cans \$3.57, pre-diced tomato cups 2 \$4.98 = \$19.51	\$0.70	Drained and rinsed beans; the teacher opens the cans
N5	No-Bake Cookie Dessert Cups	Dessert	Cookies 2 packs \$5.98, whipped topping 3 cans \$8.97, cups 1 pack \$3.99 = \$18.94	\$0.68	The same Option E as the kitchen version. No heat, so it works either way

The no-kitchen constraint card replaces items 3 and 4 with: no heat of any kind; the teacher opens every can and every sealed package; cold items travel in a cooler with ice packs and go out at minute 5, not minute 0; anything perishable that is not eaten goes in the trash at the bell, not back in a bag.

What is lost, and what is gained. The class does not demonstrate FSP 1. e) (prepare a simple recipe) with heat, and the cooking method content from Topic 6.2 does not get a second performance. Everything about the system, the math, the money, the plan, the service, the clean, and the reflection is identical, and the no-kitchen version is actually harder on the cold-chain and hold-time side, which is a real food safety lesson the kitchen version does not teach as well. Say that out loud to the class so the no-kitchen version does not read as the consolation prize.

Teacher notes on the project

- This is my own multi-kitchen system, run by the students instead of by me. My version: six costed menus for 165 students across six periods and three kitchens on a \$200 budget; an eleven-item shopping list for 112 students across four periods and four kitchens on \$150 with store-brand swaps that saved \$30.38; a one-page master flowchart in five-minute blocks marked substitute-teacher friendly; and one station card per kitchen for seven students with ingredients, ten numbered steps, safety reminders, and a “what does done look like” checklist. All four documents are in my own materials. The pedagogical move in this unit is that I planned all of it there and the students plan all of it here.
- The teacher’s job in this project is to hold the constraints and buy the food. Everything else belongs to the students. The hardest part for a teacher who has done this before is not fixing the over-budget station’s problem for them.
- Buy the day before cook day, not the same morning. Portion into four labeled station bins before the bell.
- One section or several: My own system ran the same menu across six periods on one order, which is the version that scales. If two sections cook different menus, you buy and set up twice. Decide before Lesson 6.12, because it changes which options are on the table.
- Budget reality check: \$80 per section per year for this project, plus about \$8 in paper goods if the room does not already have them. Across four sections that is about \$350. That is the single biggest line in the whole course. If the budget will only fund half, three stations instead of four with the two cheapest options brings a section to about \$55.
- The three checkpoint signatures are what keep this project from going wrong: the costing sheet signed in 6.13, the station card approved in 6.14, and the consolidated list approved in 6.15. No signature, no next step. Say that on day one.
- The reflection is the part most likely to get cut for time, and it is the part that makes this a project instead of a cooking day. Protect the eight minutes in Lesson 6.17.